



Laboratory Information System (LIS) for Medical Laboratories with Semi- Automatic Equipments

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Abstract

This proposal presents the development of a comprehensive web-based Laboratory Information System (LIS) designed for medical laboratories operating with semi-automatic testing equipment. The system addresses critical inefficiencies in current manual processes including appointment scheduling, result entry, report generation, and billing operations. The proposed solution integrates all laboratory operations from patient registration to result delivery through a centralized platform, accommodating semi-automatic equipment via manual data entry interfaces while automating downstream processes. The system architecture follows a three-tier model utilizing React.js for the presentation layer, Node.js for the application layer, and PostgreSQL for data management. The front-end is built entirely with React.js, leveraging its component-based architecture to create reusable UI elements, efficient state management, and responsive design across all devices. This approach ensures a modern user experience while minimizing development costs through open-source technologies. Key features include online appointment booking, barcode-based sample tracking, automated report generation with templates, integrated billing, real-time patient notifications via SMS/email, and comprehensive audit trails. The project will be executed over 40 weeks following a structured life cycle: Analysis (10 weeks), Design (12 weeks), Development, and Testing & Deployment (18 weeks).

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Chapter 1 : Introduction

1.1 Background of Studies

Medical laboratories form a critical component of healthcare delivery, processing diagnostic tests that inform clinical decision-making. However, many laboratories in developing regions continue to rely on manual, paper-based systems despite utilizing semi-automatic testing equipment [1]. This disconnects between testing technology and information management creates operational inefficiencies, increases error rates, and compromises patient satisfaction.

This project simulates a small to medium-sized diagnostic laboratory serving approximately 50,000 patients annually, processing 50-100 samples daily. The facility employs 5 to 8 staff members and operates with semi-automatic equipment requiring manual result transcription. Current operations utilize fragmented and decentralized systems including paper logbooks, multiple Excel spreadsheets for report generation, and standalone billing software, resulting in data silos, version control issues, and significant workflow bottlenecks [2]. Each staff member may maintain their own Excel files, leading to inconsistent data, duplication of effort, and lack of centralized patient records.

With the decision to implement React.js as the frontend technology, the proposed system will deliver a modern, responsive user experience that can adapt to various devices including desktops, tablets, and mobile phones used in laboratory settings. React's component-based architecture enables rapid development of reusable UI elements, ensuring consistency across different modules while reducing development time and maintenance effort [3]. The React ecosystem provides numerous free, open-source libraries that eliminate licensing costs while delivering enterprise-grade functionality.

The proposed system will replace the current Excel-based report generation with automated, template-driven report creation, eliminating manual formatting errors and ensuring consistent branding. By centralizing the decentralized system, all laboratory data will be stored in a single database with proper access controls, audit trails, and real-time synchronization across all staff members.

1.2 Problem Identification

The existing system operates through eight sequential manual steps:

1. Appointment scheduling via phone and paper logbook.
2. Patient registration using handwritten forms (duplicate entries across staff).
3. Manual sample collection logging in paper registers.
4. Test processing on semi-automatic equipment.
5. Manual result transcription from equipment printouts.
6. Excel-based report generation - each staff member creates reports in their own Excel templates, leading to:
 - i. Inconsistent formatting across reports.
 - ii. Version control issues (multiple copies of templates).
 - iii. Manual data entry errors.
 - iv. No centralized repository of generated reports.
 - v. Difficulty in retrieving historical reports.
7. Manual billing entry in standalone software (disconnected from patient records).

8. Phone-based result notification with physical report collection.

Critical Problems with Current Decentralized Excel-Based System:

Table 1 Problems in Existing System

Problem Area	Current Issue	Impact
Data Decentralization	Each staff member maintains separate Excel files	No single source of truth; conflicting patient records
Excel Report Generation	15-20 minutes per report; manual formatting	30-45 min total per patient; formatting inconsistencies
Version Control	Multiple Excel template versions in circulation	Reports with different formats; branding inconsistency
Data Entry Errors	Manual transcription to Excel	5-8% error rate; patient safety risks
Data Silos	Reception, lab, billing use different systems	Duplicate data entry; reconciliation challenges
Report Retrieval	Reports stored in staff personal folders	Difficult to find historical reports; no audit trail
Backup Issues	No centralized backup of Excel files	Risk of data loss; no disaster recovery

Compliance	No audit trail of who modified what	Regulatory non-compliance risk
Staff Efficiency	5-8 staff spending 25-30 hours/week on manual Excel work	High operational costs; staff burnout
Patient Experience	No online access; delayed results	Poor satisfaction; competitive disadvantage

1.3 Objectives

1. Centralize all laboratory data and billing systems in one web-based platform.
2. Automate report generation for patients.
3. Enable online appointment booking for medical tests.
4. Implement automated patient notifications via SMS/email to patients.
5. Develop a cost-effective solution using open-source technologies.

Chapter 2 Literature Review

2.1 Introduction

A Laboratory Information System (LIS) is a specialized information management system designed to support laboratory workflows including patient registration, test ordering, sample tracking, result processing, reporting, and billing. LIS plays a critical role in ensuring diagnostic accuracy, traceability, and regulatory compliance in healthcare institutions.

According to the World Health Organization (WHO), strengthening laboratory systems through digital information management improves diagnostic efficiency, reduces turnaround time, and enhances patient safety [1]. Similarly, the Centers for Disease Control and Prevention (CDC) emphasizes that laboratory information systems are essential components of public health infrastructure and quality assurance programs [5].

Modern LIS platforms are frequently integrated with Hospital Information Systems (HIS) and Electronic Health Records (EHR), allowing seamless data exchange across clinical departments.

2.2 Existing Systems / Related Work

The Sri Lankan laboratory landscape presents a mix of solutions, with significant variation between major hospital laboratories and smaller independent or branch laboratories.

Nawaloka Laboratory Services

Nawaloka Hospital's main laboratory in Colombo operates with a partially automated commercial LIS implemented approximately eight years ago, customized for local requirements. However, its five or more branch laboratories and collection centers across Colombo and suburbs operate with no integration into the main system, relying entirely on Excel spreadsheets and paper-based records. Staff at these branches frequently maintain separate Excel files for the same patients, resulting in conflicting records and delayed data retrieval [2]. This branch-level fragmentation directly mirrors the operational context of the proposed system.

Durdans Hospital Laboratory Services

Durdans Hospital operates a bespoke LIS integrated with its main hospital information system, developed specifically for internal use and not commercially available. Its collection centers across the Colombo metropolitan area, however, depend on Excel-based patient registration and result tracking. Results are telephoned through to centers, manually re-entered into Excel, and then printed for patients. During high-demand periods, staff report that Excel files take two to three minutes merely to save when handling over 200 patients per day [6]. This demonstrates a critical scalability limitation of spreadsheet-driven workflows that the proposed centralized LIS will resolve.

Asiri Laboratory Services

Asiri Laboratory Services centralizes its main operations at Asiri Central and Asiri Surgical hospitals using a hospital-integrated LIS. However, its multiple branch and collection center locations operate with limited or no integration to the main system, relying on a mix of Excel-based and paper-based workflows [7]. This further

confirms the widespread pattern of disconnected branch operations in Sri Lankan laboratory services, a gap that the proposed web-based LIS directly addresses through a single centralized platform.

Independent Small and Medium Laboratories

Sri Lanka is estimated to have 150 to 200 independent diagnostic laboratories and over 500 neighborhood laboratories, employing between 2 and 15 staff members. Over 90% of these facilities rely on Excel for report generation, with manual transcription from semi-automatic equipment being the dominant workflow [8].

Common Characteristics:

- 90%+ use Excel for report generation
- Manual result transcription from semi-automatic equipment
- No centralized patient database
- Report retrieval takes 15-20 minutes on average
- High error rates (estimated 5-8%)
- No patient portal or online services

Research shows that many laboratories in developing regions continue to rely on manual processes and spreadsheet-based systems due to financial constraints [9]. Excel-based report generation remains common because of low implementation cost and use familiarity.

However, spreadsheet-driven workflows introduce several risks:

- Data duplication
- Version control problems

- Lack of centralized storage
- Increased transcription errors
- Absence of audit trails

A study on laboratory strengthening initiatives found that integrated LIS deployment significantly reduces data errors and improves workflow efficiency [10]. While commercial LIS solutions exist for large, fully automated hospitals, limited research focuses on:

- Affordable web-based LIS
- Systems tailored for semi-automatic laboratory equipment
- Solutions optimized for small-to-medium laboratories in developing countries

Most enterprise-level LIS products are costly and require extensive analyzer integration capabilities, making them unsuitable for smaller laboratories relying on manual transcription workflows.

Therefore, there is a clear need for a centralized, cost-effective, web-based LIS designed specifically for semi-automatic laboratory environments.

Chapter 3 Proposed Methodology

3.1 Flow Diagram

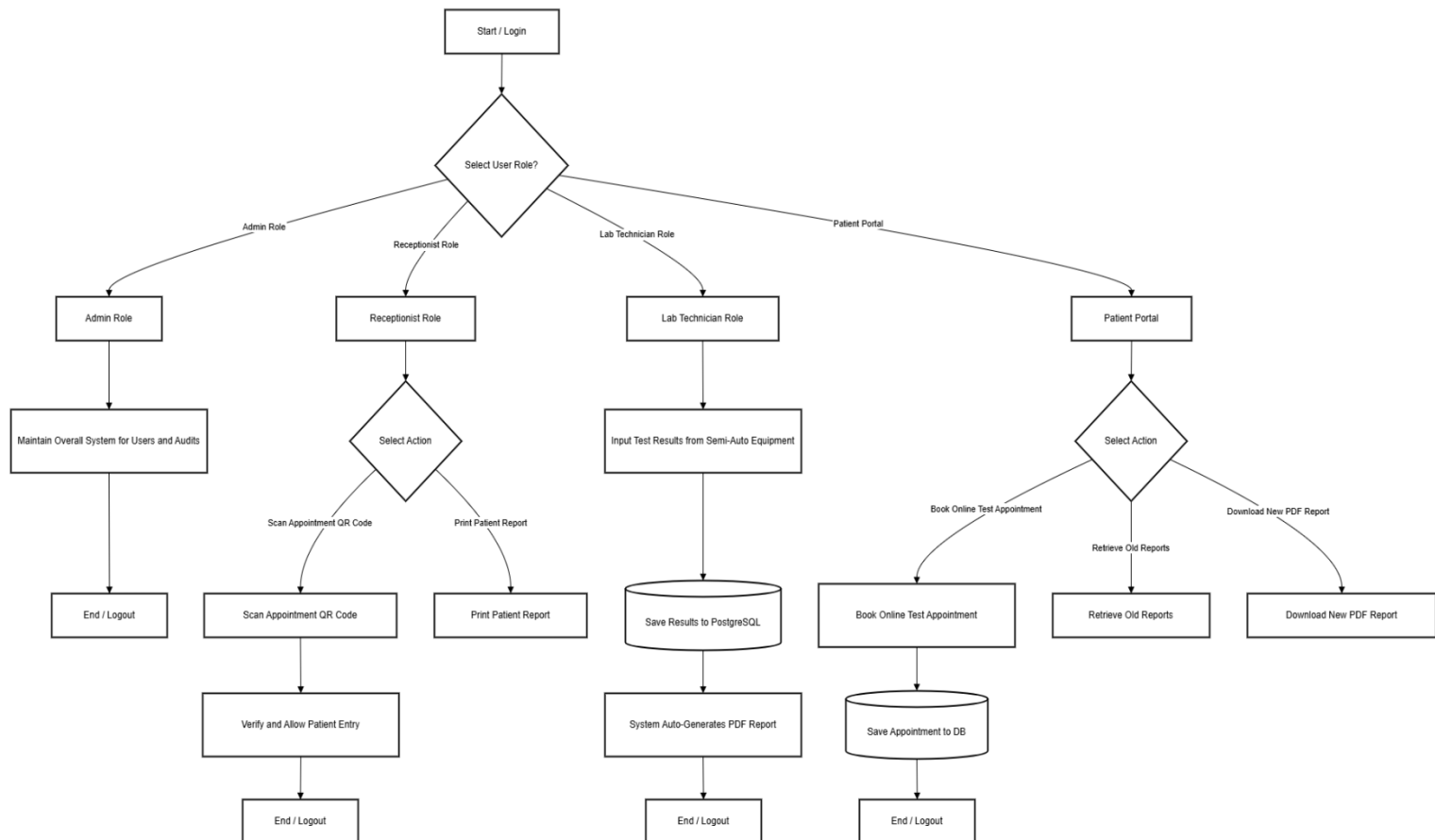


Figure 1 Flow Diagram of System

The system workflow begins with the patient booking an appointment through the online portal. On arrival, the patient is registered in the centralized database, and a barcode-labelled sample is collected. Laboratory technicians process the sample using semi-automatic equipment and manually enter the test results into the system via a structured data entry interface. The system then automatically generates a formatted PDF report using a standardized template, simultaneously triggers SMS or email notifications to the patient, and processes the billing entry. The patient may collect the report physically or access it digitally through the patient portal.

Administrators and supervisors can monitor the complete workflow in real time through the management dashboard. This end-to-end automated flow eliminates all paper-based hand-offs, manual Excel report generation, and disconnected billing processes present in the current system.

3.2 Functional Requirements

Table 2 Functional Requirements

ID	Requirement	Priority
FR-01	Online appointment booking	High
FR-02	Manual result entry for semi-automatic equipment	High
FR-03	Automated report generation with templates	High
FR-04	SMS/email notification system	High
FR-05	Role-based access control (4+ roles)	Medium

3.3 Non-Functional Requirements

Table 3 Non-Functional Requirements

Category	Specification
Security	TLS 1.3 encryption; JWT authentication; role-based access control; secure token storage; HTTPS only
Performance	Page load time < 3 seconds; support 10+ concurrent users; optimized with code splitting and lazy loading
Responsiveness	Full functionality on desktops, tablets, and mobile devices; responsive design with Tailwind CSS

Compatibility	Cross-browser support for Chrome, Firefox, Safari, Edge (latest versions)
Data	Centralized PostgreSQL database with single source of truth; 99.9% data accuracy; complete audit trail; ACID compliance

3.4 Technologies and Tools Review

Table 4 Tech stack

Component	Technology
Frontend	React.js, Tailwind CSS
Backend	Node.js
Database	PostgreSQL
Reports	React-PDF

The laboratory system can implement several cost-effective security measures to safeguard patient and operational data within the project budget. User authentication and role-based access control ensure that only authorized personnel can access sensitive information, while database encryption and SSL/TLS secure both stored and transmitted data. Regular backups, stored either locally or in the cloud, protect against accidental loss or system failures, and audit logs track user actions to maintain accountability and detect potential misuse. Optional features, such as SMS notifications, can be enabled securely without including sensitive medical details, preserving patient privacy. Additionally, keeping all software up to date prevents

vulnerabilities from being exploited. Together, these measures provide robust data protection without significant extra cost, balancing security with budget constraints.

3.5 Proposed Testing and Evaluation Method

The system will undergo a structured testing process to ensure reliability, security, and performance. Unit testing will be conducted during development to verify individual modules, followed by integration testing to validate data flow between frontend, backend, and database components. System testing will evaluate functional requirements such as appointment booking, result entry, automated report generation, billing integration, and notification services. User Acceptance Testing (UAT) will be performed with laboratory staff to ensure usability and workflow compatibility. Performance testing will assess response time and multi-user support, while security testing will verify authentication, role-based access control, and data protection mechanisms. This comprehensive evaluation approach ensures that the system meets both functional and non-functional requirements before deployment.

Chapter 4 Time Frame

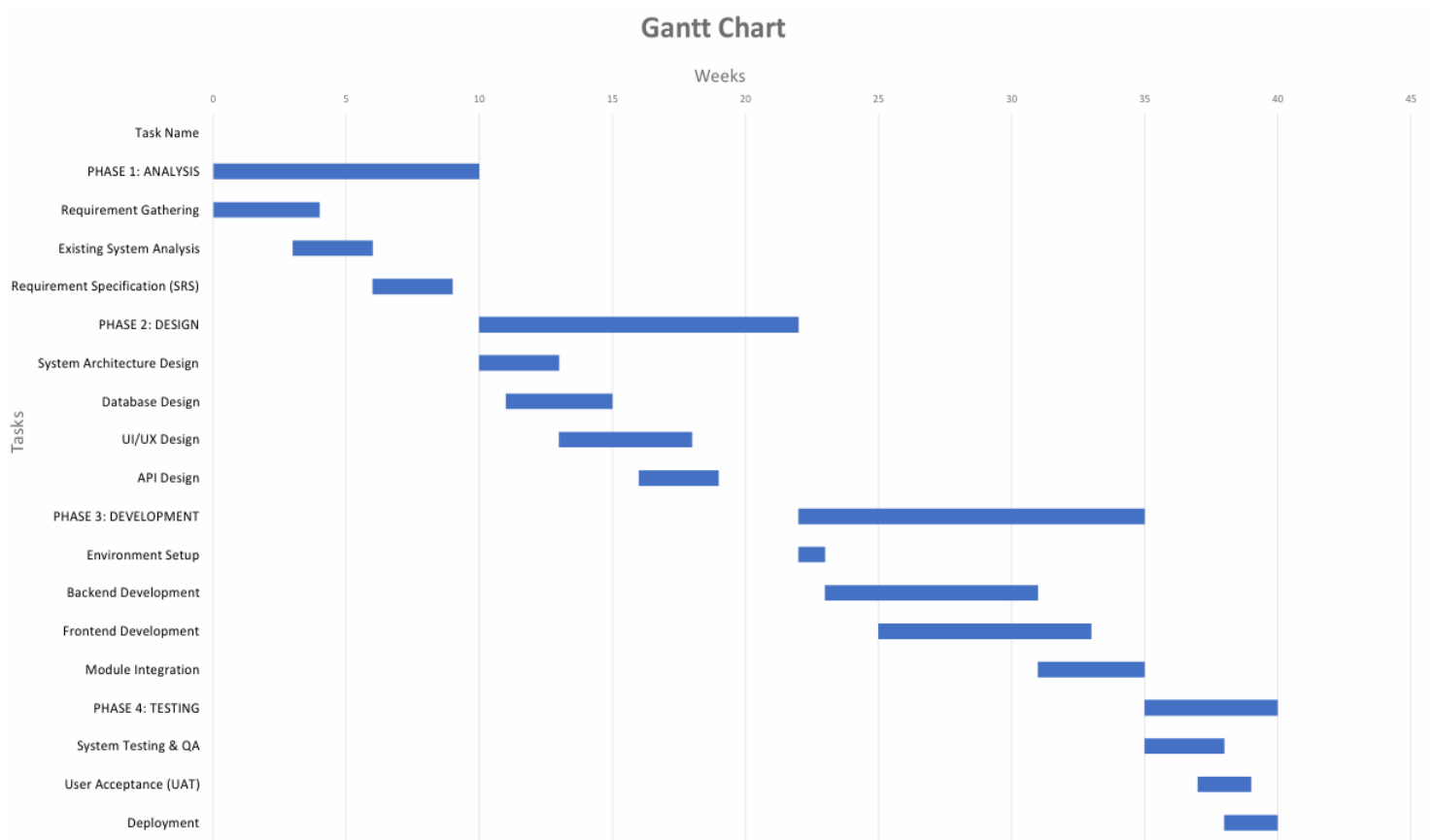


Figure 2 Gantt Chart of The System

The project timeline was developed using Microsoft Excel to create a detailed Gantt chart representing all major phases of the system development life cycle. The chart visually illustrates the structured progression of activities across Analysis (10 weeks), Design (12 weeks), Development and Testing & Deployment (18 weeks), within the planned 40-week duration. Excel was selected due to its flexibility in timeline customization, task dependency visualization, and cost-effectiveness. The Gantt chart enables effective monitoring of milestones, resource allocation, and progress tracking throughout the project lifecycle.

Chapter 5 Budget

Table 5 Cost Estimation

Item	Cost (LKR)
Domain (.com)	2,500.00
Hosting	8,000.00
Total	10,500.00

As per the client's requirement the proposed system is designed as a low-cost solution utilizing open-source technologies to minimize financial overhead. The primary expenses include domain registration (LKR 2,500) and web hosting services (LKR 8,000), resulting in a total estimated cost of LKR 10,500. Since the system is developed using open-source frameworks and tools, no additional licensing fees are required. This cost-efficient approach makes the system highly suitable for small and medium-scale medical laboratories operating within limited budgets.

The SMS notification service is an optional feature. Its activation depends on the volume of laboratory patients and the laboratory's operational priorities. Laboratories can choose to enable SMS notifications to keep patients informed about test updates, but this comes with an additional recurring cost ranging from LKR 39,000 to LKR 78,000 per year. If budget constraints or low patient volume make SMS notifications unnecessary, laboratories may opt not to activate this service, thereby avoiding the extra recurring cost.

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Appendix

Appendix 1: List of Abbreviations

The following abbreviations are used throughout this proposal document.

API - Application Programming Interface

ACID - Atomicity, Consistency, Isolation, Durability

AES - Advanced Encryption Standard

CDC - Centers for Disease Control and Prevention

CSS - Cascading Style Sheets

FR - Functional Requirement

HTTPS - Hypertext Transfer Protocol Secure

JWT - JSON Web Token

KDU - General Sir John Kotelawala Defence University

LIS - Laboratory Information System

LKR - Sri Lankan Rupee

OWASP - Open Web Application Security Project

PDF - Portable Document Format

QA - Quality Assurance

QR - Quick Response (code)

SMS - Short Message Service

TLS - Transport Layer Security

UAT - User Acceptance Testing

UI/UX - User Interface / User Experience

WHO - World Health Organization